

Yinda Chen

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Education

University of Science and Technology of China & Shanghai AI Lab — Hefei & Shanghai

Information and Communication Engineering, Ph.D. Candidate — **2024.7 ~ 2027.6 (Expected)**

- **Research Focus:**Predictive multimodal intelligence: self-supervised representation learning, embodied world models and VLA; scientific agents, foundation-model post-training, medical imaging, and connectomics
- **Advisors:**Prof. Feng Wu (CAE Academician, IEEE Fellow), Prof. Zhiwei Xiong; Connectomics research mentor: Ruobing Zhang (Fudan University)
- **Core Courses:**Algorithm Design and Analysis, Statistical Learning, Deep Learning, Reinforcement Learning
- **Honors:**NSFC Doctoral Student Project PI (awarded 2024; project period 2025–2027); Selected for the 2026 CAST Doctoral Student Program (CCF)

University of Science and Technology of China — Hefei

Computer Technology, M.S. — **2022.9 ~ 2024.7**

- **Honors:**National Graduate Scholarship (2022)
- **Research Experience:**Previously worked with Prof. Xiaoou Tang at Shanghai AI Lab in 2023

Xiamen University — Xiamen

Environmental Ecological Engineering & Economics (Dual Degree), B.S. — **2018.9 ~ 2022.7**

- **Overall Ranking: 1/31**
- **Honors:**Xiamen University Academic Star (2021), CDA Level 1 Certification (2022), Kaggle Expert
- **Advisor:**Prof. Yuanye Zhang

Research Projects

Embodied Intelligence & World Models: Pelican Series — 2026.5 ~ Present

Unified Embodied Foundation Models *Pelican-Unified 1.0 Core Contributor; Pelican-VLA 0.5 Contributor*

- In **Pelican-Unified 1.0**, core contributor to **unified VLM and action modeling**, connecting multimodal understanding, future video prediction, and low-level actions within the unified understanding, reasoning, imagination, and action framework.
- Contributed to **Pelican-VLA 0.5: Attending Before Acting Benefits Generalization**, studying **task-relevant bottleneck representations**, vision-language conditioning, and action prediction, including real-robot closed-loop validation.
- Investigated learnable **Bottleneck Tokens** for task-relevant visual information, the connection between future-state modeling and action learning, and generalization across task scenes.
- Media coverage: People's Daily, Beijing Daily / Beijing Municipal Government, Machine Heart, Zhidongxi

NSFC Young Students Basic Research Program (Doctoral Student) — 2025.1 ~ 2027.12

NSFC Project *Principal Investigator (Awarded 2024; project period 2025–2027; funding 300k RMB)*

[1] Neuroscience representations and pretraining

- Visual token construction, **autoregressive pretraining**, and multiscale representation consistency for electron microscopy and 3D medical images.
- Data-efficient **self-supervised and predictive learning** with multimodal neural imaging data.

[2] Fine-grained neuron segmentation

- **Multi-agent masking**, complementary masking consistency, and domain-adaptive segmentation for representation transfer and robustness.

[3] Connectome structure–function analysis

- Investigated neurotransmitter lateralization and its relationship to memory computation in the adult Drosophila mushroom-body connectome.
- **SfN 2026 poster abstract accepted:** Connectome-scale chemical lateralization predicts asymmetric memory computation in the adult Drosophila mushroom body.

Representation and Predictive Learning — 2022.5 ~ Present

Conference&Journal *Selected papers; author roles shown per entry*

[1] TokenUnify: Scaling Up Autoregressive Pretraining for Neuron Segmentation, ICCV 2025 **[Co-first]**

- Developed image **autoregressive pretraining with Mamba** for transferable neuron-image representations and efficient long-sequence modeling.
- Designed **mixed token-prediction objectives** to capture local structures and long-range dependencies; studied scaling behavior through theoretical analysis and neuron-segmentation experiments.

[2] MaskTwins: Dual-form Complementary Masking for Domain-Adaptive Image Segmentation
ICML 2025 **[Theory]**

- Contributed **theoretical analysis** of dual-form complementary masking from a sparse-signal reconstruction perspective, explaining its role in domain-invariant representation learning.
- Reported gains: 2.7 mIoU points on SYNTHIA to Cityscapes; 3.2 IoU points on VNCIII to Lucchi.

[3] Self-supervised Neuron Segmentation with Multi-agent Reinforcement Learning, **IJCAI 2023** **[First]**

- Developed **dbMiM**, a multi-agent reinforcement-learning framework that improves masked-image pretraining by jointly optimizing masking ratios and strategies.
- Used adaptive masking tasks to learn neuron-image representations for downstream neuron segmentation.

[4] Learning Multiscale Consistency for Self-Supervised Electron Microscopy Instance Segmentation
ICASSP 2024 **[First]**

- Combined **multiscale feature contrastive learning** and reconstruction for self-supervised EM pretraining.
- Designed a **feature-consistency objective** to address scale-dependent representation differences and evaluated the learned features for neuron instance segmentation.

[5] Unsupervised Domain Adaptation for EM Image Denoising With Invertible Networks, **IEEE TMI, 2024 (Early Access)** **[Co-author]**

- Co-authored an unsupervised domain-adaptive EM denoising method using **invertible networks** to disentangle content and noise in a domain-agnostic content space.
- Used domain alignment and pseudo-noisy image reconstruction to bridge synthetic and real noise, evaluating both restoration quality and downstream neuron segmentation.

Multimodal and Generative Models **2023.8 ~ Present**

Conference&Journal *Selected papers; author roles shown per entry*

[1] Learned Image Coding with Generative Reference of Conditional Latents
IEEE TPAMI, 2026 (accepted) **[Co-first]**

- Extended CLC to study **generative reference images** for conditional latent coding and entropy modeling.
- Integrated **local dictionaries, web retrieval, and image generation** as reference sources, with conditional latent modeling, reference synthesis, and robustness analysis.

[2] Conditional Latent Coding with Learnable Synthesized Reference for Deep Image Compression
AAAI 2025 **[Co-first]**

- Built a similarity dictionary to retrieve reference images and introduced **learnable synthesized references** for entropy-model probability estimation.
- Combined **conditional latent coding**, reference synthesis, and adaptive entropy modeling for reference-assisted learned image compression.

[3] GTGM: Generative Text-Guided 3D Vision-Language Pretraining for Medical Image Segmentation, **ICCV 2025 Workshop (VLM3D)** **[Co-first]**

- Used language-model-generated image descriptions for **3D vision-language pretraining without real paired text**.
- Combined image-text contrastive learning and generated-text guidance, with supervised downstream fine-tuning and evaluation on 10 medical-image segmentation datasets.

[4] BIMCV-R: A Landmark Dataset for 3D CT Text-Image Retrieval, **MICCAI 2024** **[First]**

- Built a 3D CT text-image retrieval dataset containing **8,069 CT volumes**, supporting retrieval between volumetric images and clinical reports.
- Developed the **MedFinder** alignment and retrieval framework for semantic search of 3D CT data.

[5] MaskFactory: Towards High-quality Synthetic Data Generation for Dichotomous Image Segmentation
NeurIPS 2024 **[Co-first]**

- Edited masks through **rigid and non-rigid deformations**, then used ControlNet to generate corresponding mask-image pairs.
- Combined deformation-driven mask editing, controlled image generation, and data-quality assessment to study synthetic training data for dichotomous image segmentation.

[6] Joint Semantic and Coded Generation for Conditional Latent Coding, **IEEE TCSVT, 2026** **[Co-author]**

- Co-authored a framework for **joint semantic and coded guidance** of reference-image generation, using text descriptions and coded features for conditional latent coding.
- Studied reference alignment, feature fusion, and probability estimation to connect controllable image generation with learned image compression.

[7] EMPOWER: Evolutionary Medical Prompt Optimization With Reinforcement Learning, **IEEE JBHI, 2025 (Early Access)** **[Co-first]**

- Combined domain knowledge and **reinforcement learning** for evolutionary medical prompt optimization, including medical attention, multi-dimensional quality assessment, evolutionary search, and semantic verification.
- The reported experiments reduced factual errors by 24.7% and improved clinician preference by 15.3% in blinded evaluation.

[8] FURINA: Free from Unmergeable Router via LINear Aggregation of mixed experts
IEEE IoTJ, 2026 (accepted) **[Co-author]**

- Co-authored a **router-free MoE-LoRA** method that linearly aggregates updates from multiple low-rank experts.
- Studied mergeable expert adaptation by integrating the aggregated updates into the base model, avoiding an inference router that prevents parameter merging.

Large Model Engineering **2023.9 ~ Present**

Projects *Core Member*

- **Foundation-model systems:** worked on image coding, foundation-model training, and LLM-driven quantitative research; used DDP and DeepSpeed for distributed training on **thousand-GPU clusters**.
- **Embodied and biomedical AI:** contributed to real-robot closed loops, Pelican-Unified pretraining, and embodied training infrastructure on **100-GPU H200** clusters; conducted medical-image and neuron-model pretraining using **64-GPU A40** environments.

Industry Internship Experience

Ubiquant Beijing 2026.5 ~ 2026.10

Quantitative Research and Large Language Models *Research Intern*

- Worked on **quantitative research** and **general large models**.
- Experienced with **thousand-GPU-scale clusters** and explored LLM applications in factor mining and data-driven decision making.
- **Core contributor to SAIL** for literature research, scientific coding, and multi-step research workflows (October 2026 technical report).
- **Contributor and co-author of IQuest-Q1**, an agentic foundation model for CLI interaction, coding, and multi-step tool use (October 2026 technical report).

Beijing Innovation Center of Humanoid Robotics (X-Humanoid) Beijing 2025.12 ~ Present

Embodied Intelligence World Model Algorithm Team *Core Member*

- Developed **unified embodied world models** for humanoid robots, connecting task-relevant perception, future-state prediction, and low-level action generation in real-robot closed loops.
- Core contributor to unified VLM and action modeling in **Pelican-Unified 1.0**, including pretraining and planning validation; contributed to task-relevant bottleneck representations and action learning in **Pelican-VLA 0.5**.
- Experienced with **100-GPU H200** cluster training for large-scale embodied foundation models.

Tencent (IEG) Shanghai 2025.8 ~ 2025.12

Qingyun Program *Research Intern*

- Worked on **game world models** and multimodal scene understanding, including **MAIN-VLA** for Game for Peace, connecting intentions, game states, and actions.
- Delivered the **Honor of Kings Lingbao** project, focusing on **TTS** and **real-time match commentary**.
- Trained and optimized Lingbao speech generation models using F5-TTS and IndexTTS2.
- Co-authored **FURINA**, a router-free MoE-LoRA method that linearly aggregates low-rank expert updates for merging into the base model (IEEE IoTJ, accepted 2026).

PLA General Hospital (301 Hospital) Beijing 2023.9 ~ 2024.2

Data Compression Group *Research Intern*

- Collaborated with Academician Qionghai Dai's team on efficient data compression research.
- Studied CT and MRI compression through content representations, reference images, and efficient coding.

Academic Internship Experience

Imperial College London London (Remote) 2022.11 ~ 2023.8

Data Science Institute *RA*

- Worked with Rossella Arcucci and Che Liu on 3D medical multimodal learning, including generated-text-guided pretraining (GTGM) and CT text-image retrieval (BIMCV-R / MedFinder).
- Developed image-text contrastive learning and retrieval methods; contributed to BIMCV-R, a dataset containing **8,069 CT volumes**.

Xiamen University WISER Club Xiamen 2021.8 ~ 2022.7

Data Mining Group *Academic Community Member*

- Responsible for designing and discussing data mining courses, teaching clustering and Transformer sections.
- Mentored undergraduate machine-learning projects and organized campus competitions.

Wang Yanan Institute for Studies in Economics
Xiamen University Xiamen 2020.8 ~ 2021.12

Econometrics *Research Assistant*

- Assisted Associate Professor Jiong Zhu in completing territorial economic statistics, conducting visual feature extraction of homestead information.
- Developed satellite-image analysis tools and extracted visual features to study land-use changes.

Honors and Awards

- **Selected for the 2026 CAST Doctoral Student Program (CCF)** **2026.10**
  - Cultivating society: China Computer Federation (CCF)
- **Zenghua Scholarship, University of Science and Technology of China** **2026**
- **Interdisciplinary Contest in Modeling (ICM)** **2024**
  - Outstanding Winner
- **NSFC Young Students Basic Research Program (Doctoral Student)** **2024.12**
  - Principal Investigator; awarded 2024, project period 2025–2027, funding RMB 300,000
- **National Graduate Scholarship** **2022.12**
  - University of Science and Technology of China

- **Xiamen University Academic Star** **2021.12**
  - Undergraduate research honor
- **'Jingrun Cup' Mathematics Competition (Mathematics Majors)** **2021.10**
  - First Place, Xiamen University
- **'Internet+' Innovation Competition** **2021.08**
  - Gold Award, Fujian Province
- **National College Student Mathematics Competition Finals (Non-Mathematics)** **2021.05**
  - National Second Prize
- **'Challenge Cup' Academic Competition** **2021.05**
  - First Prize, Fujian Province
- **National College Student Mathematics Competition (Non-Mathematics)** **2020.11**
  - First Place, Fujian Province

Technical Skills

- **Programming**: Python, C, C++, Java, MATLAB, LaTeX
- **Deep Learning**: PyTorch, TensorFlow, DeepSpeed, DDP
- **Languages**: English (TOEFL 110, GRE 328), Chinese (Native)
- **Tools**: Git, Docker, CUDA, HPC

Service: Reviewer for CVPR'25, ICCV'25, WACV'26, NeurIPS'24, ICML'25, ICLR'24, MICCAI'25, ACM MM'24, AISTATS'24, IJCV, and TIP.

Invited Talk: Vision-Language Pretraining Using Generative Methods, JD AI Team, 2023.03.